HARIHARAN T S

AI / ML Engineer | Generative AI Developer | LLM Applications

hari04022005@gmail.com | +91 6382939578 | linkedin.com/in/hariharan-ts | github.com/Tenyson-S | hariharan-ts-portfolio.vercel.app

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and Computer Science fresher (BE CSE — AI & ML, CGPA 8.79) with proven experience designing and deploying end-to-end machine learning systems. Built a production-grade AI finance application achieving 91% transaction classification accuracy at sub-200ms inference latency using TensorFlow, FastAPI, and NLP pipelines. Hands-on with Prompt Engineering, RAG pipelines, LangChain, and LLM API integration. Proficient in Python, PyTorch, Scikit-learn, Hugging Face Transformers, and full-stack deployment. Solved 400+ LeetCode problems demonstrating strong algorithmic foundations. Seeking AI Engineer, GenAI Developer, or LLM Engineer roles where I can contribute to production AI systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript

AI / ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Keras, CNNs, Deep Learning, Neural Networks

Generative AI & LLM: LangChain, Prompt Engineering, RAG Pipelines, LLM API Integration (OpenAI / Gemini), Hugging Face Transformers, Vector Embeddings, ChromaDB, FAISS, Semantic Search

NLP & Computer Vision: NLP, Text Classification, Named Entity Recognition, OCR (Tesseract), OpenCV, Image Classification

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Augmentation, Model Evaluation (Precision, Recall, F1, Confusion Matrix), Feature Engineering

Backend Development: FastAPI, REST API Design, Async I/O, Microservices Architecture, MySQL, SQLite, Firebase

Frontend & Deployment: React.js, JavaScript, Context API, Streamlit, Vercel, Docker (basic)

Tools & Platforms: Git, GitHub, Postman, VS Code, Jupyter Notebook, Google Colab

Core CS: Data Structures & Algorithms (400+ LeetCode), Object-Oriented Programming, System Design

PROJECTS

AI-Powered Personal Finance Management System — *2025 – Present*

Tech Stack: *Python · FastAPI · TensorFlow · Scikit-learn · Tesseract OCR · NLP · LangChain · Prompt Engineering · SQLite · Firebase · React.js*

- Architected an end-to-end AI pipeline (SMS/receipt ingestion → OCR → NLP preprocessing → multi-class ML classification → REST API response) achieving 91% transaction classification accuracy with <200ms inference latency on live data.
- Trained and optimized a multi-class deep learning classifier on 5,000+ labeled financial transactions using TensorFlow and Scikit-learn; applied data augmentation techniques to resolve class imbalance, improving minority-class F1-score by ~18%.
- Built scalable async backend with 12+ FastAPI endpoints and decoupled ML microservices; implemented offline-first SQLite + Firebase sync architecture supporting 10,000+ records with <50ms DB read latency.
- Integrated a prompt-engineered LLM insight layer using LangChain and Gemini API to generate natural language budget summaries and spending recommendations, reducing manual financial analysis time by ~80%.
- Deployed full-stack application (React.js frontend + FastAPI backend) with modular, CI-ready architecture; system handles simultaneous multi-source ingestion from SMS and scanned receipts.

CNN-Based Traffic Sign Recognition System — *2024*

Tech Stack: *Python · TensorFlow · PyTorch · CNN · OpenCV · Scikit-learn · GTSRB Dataset*

- Designed and trained a deep Convolutional Neural Network (CNN) on the GTSRB dataset (50,000+ images, 43 classes) achieving 96.4% test accuracy; benchmarked against baseline models to validate performance gains.
- Addressed class imbalance using class-weighted cross-entropy loss and offline augmentation (rotation, zoom, brightness shift); reduced misclassification rate by 22% on underrepresented sign categories.

- Evaluated model robustness using precision, recall, F1-score, and confusion matrix across all 43 classes; identified and documented top failure modes for safety-critical sign categories.
- Optimized OpenCV preprocessing pipeline for deployment readiness, achieving 30+ FPS inference throughput on CPU for real-time detection scenarios.

Document Q&A Chatbot — RAG Pipeline *2025*

Tech Stack: *Python · LangChain · Google Gemini API · ChromaDB · Hugging Face Transformers · Sentence Embeddings · Streamlit*

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline enabling natural language Q&A over custom PDF documents using LangChain orchestration and Google Gemini as the LLM backend.
- Implemented semantic document chunking, Hugging Face sentence-transformer embeddings, and ChromaDB vector store, achieving 89% top-3 retrieval accuracy on a custom eval set of 200 Q&A pairs.
- Designed a Streamlit-based conversational interface with multi-turn memory and source citation; optimized context window usage via dynamic chunk scoring, reducing hallucination rate by ~30%.

EDUCATION

B.E. Computer Science & Engineering (AI & ML) *2022 – 2026*

Sethu Institute of Technology, Kariapatti | CGPA: 8.79 | Honors: Full Stack Development

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Database Management Systems

CERTIFICATIONS

- AI Master Class (30-day Hands-on Program) — Novitech Pvt Ltd
- AI Primer Certification — Infosys Springboard
- Project-Based Robotics Training (2 programs) — Zetspire Technologies

ACHIEVEMENTS & AWARDS

- Department Topper — Awarded Academic Star & Above and Beyond honors for sustained academic excellence (CGPA 8.79).
- Winner in 7+ Inter-college Technical Symposiums — demonstrated technical depth and problem-solving across competitive events.
- Top 5 Finalist — Hackathon at Mohammed Sathak Engineering College.
- 400+ Data Structures & Algorithms problems solved on LeetCode across arrays, strings, trees, graphs, and dynamic programming.